MILITARY SPECIFICATION

CABLE REEL MARK 1 MOD 1

(FOR FIRE CONTROL SYSTEM MARK 101)

This specification has been approved by the Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. - This specification establishes the requirements for the procurement of a cable reel, BUWEPS LD 161945, and the necessary adapters to enable its use with the Bearing Transmitter Mk 17 Mods 0-10.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, drawings and publications, or such portion thereof as may be designated herein, of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes, Wood, Cleated-Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner
PPP-B-636	Boxes, Fiber

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing; Aeronautical and Associated Equipment, General Specification for
MIL-S-901	Shockproof Equipment, Class HI (High Impact) Shipboard Application, Tests for

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:20Z
Check the source to verify that this is the current version before use.

MIL-L-10547	Liners, Case, Waterproof
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of
MIL-F-18870	Fire Control, Optical and Associated Equipment Intended for Shipboard Use, the General Specifications for Design and Procurement of
MIL-W-21927	Weapons, Handling and Preparation for Delivery of; General Requirements for

STANDARDS

MIL-STD-108	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibration of Shipboard Equipment

DRAWINGS

NAVY DEPARTMENT

BUWEPS LD 161945	Cable Reel, Mk 1 Mod 1, and all parts listed thereon

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

NAVWEPS OD 10218	Cable Reel Mk 1 Mod 1 Factory Acceptance Tests

(Copies of specifications, standards, drawings and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:20Z
Check the source to verify that this is the current version before use.

2.2 Other Publications. - The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation to bids shall apply.

NATIONAL BUREAU OF STANDARDS PUBLICATIONS

Handbook H28	Screw Thread Standards for the Federal Service

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D. C.)

CONSOLIDATED CLASSIFICATION COMMITTEE PUBLICATION

Consolidated Freight Classification Rules

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Illinois.)

AMERICAN TRUCKING ASSOCIATION, INC. PUBLICATION

Motor Freight Classification Rules

(Application for copies should be addressed to the Issuing Officer, American Trucking Association, Inc., 1424 16th St., N.W., Washington 6, D. C.)

AMERICAN STANDARDS ASSOCIATION

Z 24.3(1944)	American Standards for Sound Level Meters
Z 24.7(1950)	American Standard Test Code for Noise Measurement
Z 24.10	American Standards Association - Sound Levels

(Application for copies should be addressed to the American Society of Engineers, 29 W. 39th St., New York, N. Y.)

3. REQUIREMENTS

3.1 Description. - The cable reel is a device that pays out electrical ribbon cable to the Bearing Transmitter Mk 17 Mods 0-10 as the periscope is lowered and winds the cable in as the periscope is raised. The cable reel can be located directly above the bearing transmitter, or remotely, in which case a pulley shall be located directly above the bearing transmitter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:20Z
Check the source to verify that this is the current version before use.

3.2 General Requirements

3.2.1 Pre-production samples. - Unless otherwise specified in the contract or order, pre-production samples of the cable reel shall be manufactured using the methods and procedures proposed for the production lot. These samples will be tested as specified in section 4 herein, and are for the purpose of determining that the contractor, prior to his starting production, is capable of producing items that comply with the technical requirements of the contract. Provision of samples shall be in accordance with 4.1.2.

3.2.2 Materials. - Unless otherwise specified in the contract, requisition or order, all materials and purchased parts entering into the final assembly of the cable reel shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in BUWEPS LD 161945.

3.2.3 Finishes, Coatings, Constructions and Dimensions. - Finishes, coatings, construction and dimensions for the cable reel shall be strictly in accordance with the applicable drawings, specifications, and publications listed in BUWEPS LD 161945.

3.2.4 Moisture, Fungus and Corrosion. - The cable reel shall be rendered moisture, fungus and corrosion resistant in accordance with detail requirements specified in the drawings and specifications listed in BUWEPS LD 161945.

3.2.5 Marking. - Marking of assemblies, sub-assemblies and parts shall be in accordance with the applicable drawings listed on BUWEPS LD 161945.

3.2.6 Interchangeability. - Component parts of assemblies fabricated in agreement with their applicable drawings and specifications shall be interchangeable with mating parts of the unit and other like units. If deviation greater or smaller than the tolerances shown on drawings or specifications are required for interchangeability, or if hand fitting beyond that shown on applicable drawings is required, then changes in dimensions are necessary and revision and correction of the applicable drawings are required. (see 3.2.9)

3.2.7 Threads. - Unless otherwise specified, all threads shall be in accordance with the National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings, and the NO-GO gaging practice shall be as set forth in the aforementioned Handbook H28.

3.2.8 Request for Deviation. - Each request that a deviation be authorized shall have been submitted to the Procuring Activity, and the deviation shall have been authorized by the Bureau of Naval Weapons or by a Naval Activity delegated such authority, prior to the production of the affected item, or prior to the performance of a particular operation of the item. The deviation shall apply only to this specific item or operation. The request shall define the suggested deviation in sufficient detail to permit the Procuring Activity to evaluate its effect on performance and reliability, and the reasons for the suggested deviation shall be stated. Each request that a deviation be authorized, and the authorization, shall state specifically the extent of the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:20Z
Check the source to verify that this is the current version before use.

departure, the identifying number of the item, the contractor (or bidder) and the contract number. A deviation is essentially temporary in character, and usually represents a departure limited to a specific period, to a specific number of items, or to the contract to which it applies. It may require an equitable reduction in contract price if a decrease in the cost of performance results therefrom.

3.2.9 Conflicting Requirements. - Conflicting requirements arising between this specification, publications or drawings listed herein shall be referred in writing to the Procuring Agency or appointed agent for interpretation and clarification.

3.3 Performance Requirements and Product Characteristics. - The cable reel shall meet the following performance requirements and product characteristics:

3.3.1 The cable reel shall pass the following tests specified in NAVWEPS OD 10218:

3.3.1.1 Visual inspection for quality workmanship.

3.3.1.2 Spring motor tension tests of 11 lbs minimum for a fully reeled in cable, and 17 lbs maximum for a fully extended cable.

3.3.1.3 Electrical continuity tests for individual conductors.

3.3.1.4 Dielectric strength test of 1000 volts AC for one (1) minute.

3.3.1.5 Insulation resistance test with minimum resistance of 500,000 ohms.

3.3.2 Mechanical Noise. - The cable reel shall not generate noise such that the arithmetical average of the sound pressure levels in the four octave bands 300-600, 600-1200, 1200-2400, and 2400-4800 cycles per second exceeds 65 decibels.

3.3.3 Endurance. - The cable reel shall be capable of withstanding the endurance test specified in 4.5.7 and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4 Environmental Requirements (See 6.3)

3.4.1 Enclosure. - The cable reel shall be capable of withstanding the drip-proof test specified in 4.5.1 and thereafter shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.2 Vibration

3.4.2.1 Transportation. - The cable reel shall be capable of withstanding the vibration tests specified in 4.5.2.1 and thereafter show no damage and shall meet all the requirements of section 3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:20Z
Check the source to verify that this is the current version before use.

3.4.2.2 Operating. - The cable reel, less pack and packaging, shall be capable of withstanding the vibration tests specified in 4.5.2.2. During and after the tests, the cable reel shall show no damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.3 Shock

3.4.3.1 Transportation, Storage and Handling. - The cable reel shall be capable of withstanding the shock tests specified in 4.5.3.1 and thereafter shall show no damage and shall meet all the requirements of section 3.

3.4.3.2 Operating. - The cable reel, mounted as specified in 4.5.3.2, shall be capable of withstanding shock tests as specified in 4.5.3.2. During and after the test, the cable reel shall show no damage and shall meet all performance requirements and product characterisitics of this specification.

3.4.4 Climatic Condition. - The cable reel shall be capable of withstanding the climatic conditions specified in 4.5.4 and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

3.4.5 Humidity and Fungus. - The cable reel shall be capable of withstanding humidity and fungus tests specified in 4.5.6, and thereafter shall show no corrosion or damage and shall meet all the performance requirements and product characteristics of this specification.

3.5 Clean-up. - Prior to and after final assembly, all parts, components and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. Burrs and sharp edges as well as rosin flash shall be removed.

3.6 Workmanship. - The cable reel, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, riveting, machine screw assemblage, welding, brazing, and freedom of parts from burrs and sharp edges.

3.6.1 The workmanship exhibited in the pre-production sample will be evaluated to determine acceptability. The approved standards of workmanship will thereby become a minimum production requirement for all units offered for acceptance.

4. QUALITY ASSURANCE PROVISIONS

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:20Z
Check the source to verify that this is the current version before use.

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Gages. - The contractor shall provide himself with whatever gages are necessary and adequate to insure that the material to which this specification applies will meet the dimensional requirements shown on the applicable drawings. If so stipulated in the contract, order, or requisition, the Government will furnish drawings of pertinent Navy final inspection gages for guidance in the design of the contractor's inspection gages. However, such procedure shall not serve to relieve the contractor of his responsibility, in the design and manufacture of such gages as may be required in the satisfactory fulfillment of the contract requirements, but is intended to facilitate acceptance of all components and assemblies by Navy final inspection gages.

4.1.2 Pre-production Sample. - Unless otherwise specified for each contract or order, a representative sample of five (5) cable reels shall be submitted to the Government inspector, preserved, packaged and packed as specified in the contract or order, for pre-production tests at an activity designated in the contract or by the Bureau of Naval Weapons. Further production of the cable reel by the contractor prior to approval of the contracting activity of the pre-production samples shall be at the contractor's risk. Pre-production samples accepted will be applied as part of the quantity specified by the contract or order.

4.1.3 Periodic Production Samples. - The tests of 4.4 and 4.5 shall be evaluated as follows:

Step 1. - Upon initiation of production, select one (1) unit from each group of ten (10) units for submission to an activity designated by the procuring agency.

Step 2. - Upon satisfactory compliance of each unit from three (3) consecutive groups, select one (1) unit from each group of twenty-five (25) units.

Step 3. - Upon satisfactory compliance of each unit from three (3) consecutive groups, select one (1) unit from each group of fifty (50) units. Continue this procedure until a defective unit is found.

Step 4. - When a defective unit is found in any group, the group represented by the sample shall be rejected. Shift to Step 1 and proceed as indicated. Disposition of groups represented by failing samples shall be made by the Government.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:20Z
Check the source to verify that this is the current version before use.

4.2 Classification of Tests. - The inspection and testing of the cable reel shall be classified as follows:

4.2.1 Acceptance Tests. - These tests are to be accomplished on each cable reel being submitted for acceptance under contract. These tests are defined in 4.4.

4.2.2 Pre-Production and Periodic Production Tests. - Pre-production and periodic production tests are those which are accomplished on samples selected as specified in 4.1.2 and 4.1.3. These tests are detailed in 4.5.

4.3 Test Equipment. - The following items of test equipment are required to perform the acceptance tests set forth in this specification:

4.3.1 One (1) fish-type weighing scale (0-25 lbs range).

4.3.2 One (1) ohmmeter (zero to infinity range).

4.3.3 One (1) high potential tester (1000 volt range).

4.3.4 One (1) megohmmeter (500 volt range).

4.4 Acceptance Tests

4.4.1 The contractor shall conduct acceptance tests as specified in 4.2.1 to assure that the cable reel is in compliance with the requirements of 3.2, 3.3.1, 3.5 and 3.6.

4.4.2 Test Conditions. - Unless otherwise specified the cable reel shall be subjected to acceptance tests under the following conditions:

4.4.2.1 Temperature: Room ambient 65 degrees F to 95 degrees (18.33 degrees C to 350 degrees C).

4.4.2.2 Altitude: Normal ground.

4.4.2.3 Vibration: None.

4.4.2.4 Humidity: Room ambient to 95 per cent relative maximum (except dielectric test for which specific humidity is specified).

4.4.3 Test and Inspection Equipment Facilities. - The manufacturer shall furnish and maintain all necessary test equipment, facilities and personnel for performing all acceptance tests. The test equipment shall be adequate in quantity, and when definite requirements are not specified, they shall be of sufficient accuracy and quality to permit performance of the required acceptance tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:20Z
Check the source to verify that this is the current version before use.

4.4.4 Test Procedure. - Acceptance tests shall be performed in accordance with NAVWEPS OD 10218.

4.4.4.1 Preservation, Packing, and Packaging. - Preservation, packing, and packaging of the cable reel shall conform to section 5 of this specification.

4.5 Pre-Production and Periodic Production Tests

4.5.1 Enclosure. - The cable reel shall be subjected to drip-proof tests in accordance with procedures specified in Standard MIL-STD-108.

4.5.2 Vibration

4.5.2.1 Transportation. - The cable reel, packaged and packed as specified in section 5., shall be vibration tested as specified in Specification MIL-W-21927, section 4.3.

4.5.2.2 Operating

4.5.2.2.1 Mounting. - The cable reel shall be mounted on the vibration testing machine to simulate as realistically as possible the service installation including vibration isolator or mounting devices. The cable reel shall be secured to the vibration testing machine in such a manner that the direction of vibration will be along each of the principle axes separately. The cable reel may be reoriented on the horizontal vibrating machine between each test, if necessary, to obtain such motion. At no time shall the cable reel be mounted in any other way than its normal position.

4.5.2.2.2 Test Procedures. - The cable reel shall be subjected to the vibration tests specified in Standard MIL-STD-167, Type 1.

4.5.3 Shock

4.5.3.1 Transportation, Storage and Handling. - The cable reel, packed and packaged as defined in 5., shall be subjected to the appropriate rough handling tests specified in Specification MIL-W-21927, section 4.3. Selection of tests shall be in accordance with section 4.2 of MIL-W-21927.

4.5.3.2 Operating. - The cable reel shall be mounted on the standard mounting bracket of the shock testing machine such as to simulate as realistically as possible the service installation. The cable reel shall be subjected to the shock tests specified in Specification MIL-S-901.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:20Z
Check the source to verify that this is the current version before use.

4.5.4 Climatic Conditions. - The cable reel shall be tested for satisfactory operation at the ambient temperatures and conditions, and cold weather operation conditions as specified herein. When separate high temperature and low temperature chambers are not available, it is permissible to perform the hot and cold tests in the same chamber by raising the ambient air temperature from the low to the high temperature as specified in 4.5.4.2.

4.5.4.1 The cable reel shall be chilled to zero degrees F with power off and covers in place. When the air temperature inside the cable reel has dropped to within five degrees of the specified minimum temperature, the cable reel shall be allowed to stand for a minimum period of 10 hours. At the end of this period, and while still at the test temperature, the cable reel shall meet the performance and other requirements of this specification.

4.5.4.2 After the tests described in 4.5.4.1, the ambient air temperature shall be raised to plus 110 degrees F with relative humidity between ninety-five percent (95%) and ninety-eight percent (98%) the covers of the cable reel on, and the cable reel allowed to stand in a "ready" condition. The ambient air temperature rise shall not exceed 10 degrees per hour with a maximum of 30 degrees in 8 hours. The cable reel shall operate satisfactorily at any time during this changing of the ambient air temperature. After 10 hours at 110 degrees, the cable reel shall be operated continuously for a minimum period of 50 hours under the conditions described and then tested for a satisfactory operation under these same conditions.

4.5.5 Mechanical Noise. - The cable reel shall be tested in accordance with test procedures described in the current American Standard Test Code for noise measurement, Z24.7 (1950), except that measurements shall be made with both a sound level meter conforming to American Standard Z24.3 (1944) and an octave band filter set conforming to American Standard Z24.10. At each microphone station, an octave band analysis of the noise shall be made.

4.5.6 Humidity and Fungus.

4.5.6.1 The cable reel shall be tested for resistance to humidity and fungus damage in accordance with the test specified in Specification MIL-E-5272.

4.5.6.2 Upon completion of the humidity and fungus tests, the cable reel shall be removed from the exposure chamber and operated for 24 hours. The cable reel shall then pass the dielectric and insulation resistance tests specified in 3.3.1.

4.5.7 Endurance. - After completion of all other type tests, the cable reel shall be operated and periodically tested for satisfactory operation until a total of 750 hours operating time has elapsed (including operating time elapsed during the performance of the tests under 4.5.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:20Z
Check the source to verify that this is the current version before use.

through 4.5.6) and then shall again pass the test problems and operate satisfactorily. During this period the cable shall be extended for a distance of 23 feet, then allowed to retract in the cable reel a total of 5000 times. Test conditions during the endurance test may be any condition that is specified in the environmental tests.

4.6 Failure of the cable reel to meet any of the requirements and tests of this specification shall be considered cause for rejection.

4.7 Waiver of Tests. - Tests specified in 4.5.2.1 and 4.5.3.1 may be waived by the procuring agency for Level C packs only, provided the contractor guarantees safe delivery to the using activity.

4.8 Functional Tests. - At the conclusion of the tests specified in 4.5.2.1 and 4.5.3.1, functional and other tests, as applicable, shall be conducted on the cable reel to determine freedom from operational malfunction and compliance with all the requirements of section 3.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.

5.1.1 Level A. - In addition to the following, Level A packaging shall be in accordance with Specification MIL-P-17555.

5.1.1.1 Cleaning, Drying and Preservative Application. - Not applicable.

5.1.1.2 Unit Packaging. - Unit packaging of the cable reel shall be in accordance with Method IIb of Specification MIL-P-116. The unit package shall contain one (1) cable reel and connecting box packaged in a fiberboard box in accordance with Specification PPP-B-636, class 2.

5.1.1.3 Intermediate Package. - Not applicable.

5.1.1.4 Cushioning. - The unit package, and pack, shall incorporate sufficient cushioning material, bracing, or other adequate shock absorbing devices, to ensure that the cable reel will meet all the requirements of section 3 after being tested in accordance with 4.5.2.1 and 4.5.3.1.

At the time of packaging or packing, the moisture content of the cushioning material shall be no more than 5%. The pH range of the material shall be between 6.0 and 8.0.

5.1.2 Level B. - Not applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:20Z
Check the source to verify that this is the current version before use.

5.1.3 Level C.

5.1.3.1 Cleaning, Drying and Preservative Application. - Not applicable.

5.1.3.2 Unit Packaging. - Unit packaging of the cable reel shall be in accordance with Method III of Specification MIL-P-116. The unit package shall contain one (1) cable reel and connecting box.

5.2 Packing.

5.2.1 Level A. - In addition to the following, Level A packing shall be in accordance with Specification MIL-P-17555.

5.2.1.1 Exterior Containers. - One (1) cable reel packaged as described in 5.1.1.2 shall be packed in a wood box in accordance with Specification PPP-B-601 overseas type or PPP-B-621, class 2.

5.2.2 Level B. - In addition to the following, Level B packing shall be in accordance with Specification MIL-P-17555.

5.2.2.1 Exterior Containers. - One (1) cable reel packaged as described in 5.1.1.2 shall be packed in a wood box in accordance with Specification PPP-B-601, domestic type or PPP-B-621, class 1.

5.2.3 Level C. - The cable reel packaged in accordance with 5.1.3.2 shall be packed to afford protection against damage during shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking

5.3.1 Special Marking. - None, unless otherwise specified.

5.3.2 Normal Marking. - In addition to the markings required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129, including serial numbers.

6. NOTES

6.1 Intended Use. - The Cable Reel Mark 1 Mod 1 is intended to be used with Bearing Transmitter Mk 17 Mods 0-10 in Fire Control System Mk 101.

6.2 Ordering Data. - Procurement documents should specify the specific title, number, and date of this specification and other documents. Section 5 should be adhered to for selection of applicable levels of preservation, packaging, and packing. Selection of Level C shall not be made unless immediate use is definitely known.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:20Z
Check the source to verify that this is the current version before use.

6.2.1 The attention of the contracting officer is invited to the Quality Assurance Provisions and options in Specification MIL-P-116, and to paragraph 4.8 of this document.

6.2.2 Criteria for use of proper level of preservation, packaging and packing shall be as follows:

For Level A. - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level B. - This level shall be used only when it is definitely known that the item will be held in covered storage, either in domestic or overseas locations, for a period of 6 months or less.

For Level C. - This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

6.3 These requirements are included in this specification as design objectives only. As of the date of this specification, the applicable tests listed in 4.5 herein have not been conducted to verify that the Cable Reel Mk 1 Mod 1 meets these requirements.

Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T17:20Z
Check the source to verify that this is the current version before use.